

Clustering and Initialization: k-means vs. k-means++

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This project studies whether k-means++ initialization makes k-means more stable than random initialization. I implemented k-means from scratch and ran 50 trials on separated synthetic data, overlapping synthetic data, and the Wine Quality dataset. The main measures are inertia and the standard deviation of inertia.

1. Introduction

Clustering is an unsupervised learning task. The goal is to find groups in data without using labels during training. K-means is one of the simplest clustering algorithms. It represents each group by a centroid and assigns every point to the nearest centroid.

The problem is that k-means can converge to different local minima. This happens because the k-means objective is not convex. If the initial centers are bad, the final clustering can also be bad. The aim of this project is to compare simple random initialization with k-means++ initialization.

2. Theory

The k-means objective is the sum of squared distances from each point to its closest center. This value is called inertia. Lloyd's algorithm repeats two steps: assign each point to the nearest center, then update each center as the mean of its assigned points.

Random initialization chooses K data points as the first centers. K-means++ is more careful. It chooses the first center randomly, then chooses later centers with probability proportional to $D(x)^2$, where $D(x)$ is the distance to the closest center already chosen. This tries to spread the starting centers over the data.

3. Methodology

I wrote the distance calculation, random initialization, k-means++ initialization, assignment step, centroid update, convergence check, and inertia calculation. I did not use a clustering library.

The Wine Quality data are loaded from three local files in /Users/shynggys.babir/Desktop/Data: winequality-red.csv, winequality-white.csv, and winequality.names. The red and white CSV files are joined into one dataframe named df. The 11 physicochemical columns are used for clustering, while quality and wine_type are not used for training. The features are standardized because they have different scales.

Dataset	Samples	Features	K	Reason
Separated synthetic	600	2	4	easy clear clusters
Overlapping synthetic	600	2	4	harder overlapping clusters
Wine Quality	6,497	11	2	real high-dimensional data

Table 1. Datasets used in the project.

For every dataset and every method I ran 50 trials. The maximum number of iterations was 100, and the tolerance was 0.0001.

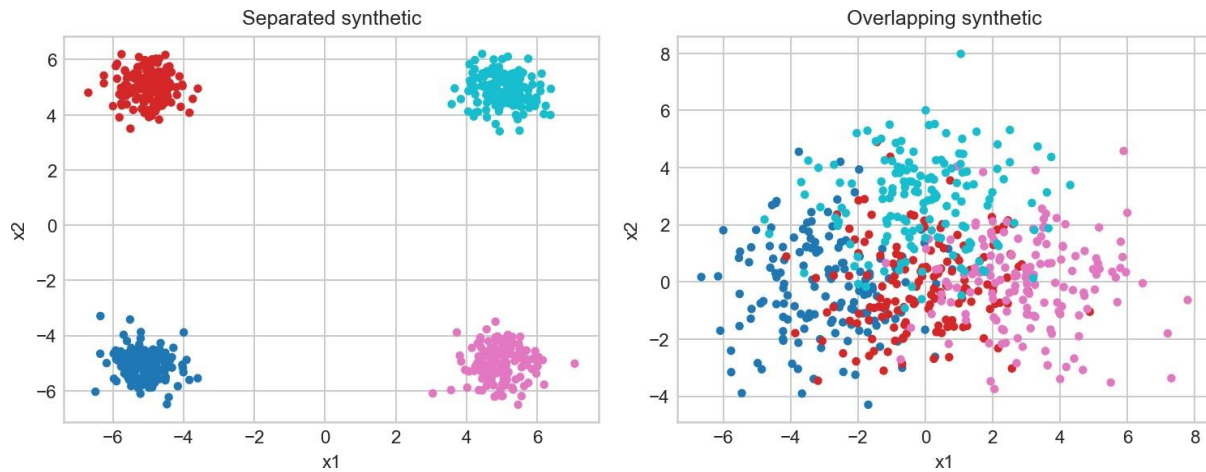


Figure 1. Synthetic datasets.

4. Results

Table 2 gives the inertia statistics. Lower mean inertia is better. Lower standard deviation means the method is more stable across repeated runs.

Dataset	Method	Mean	Std.	Min	Max
Separated synthetic	random	2,614.23	3,482.61	357.31	7,918.24
Separated synthetic	kmeans++	357.31	0.00	357.31	357.31
Overlapping synthetic	random	2,020.49	46.88	2,011.01	2,252.77
Overlapping synthetic	kmeans++	2,021.11	49.92	2,011.01	2,265.92
Wine Quality	random	57,578.82	1,471.06	56,143.92	60,530.09
Wine Quality	kmeans++	57,128.81	1,430.20	56,143.92	60,530.09

Table 2. Inertia statistics over 50 trials.

Dataset	Mean reduction	Std. reduction
Separated synthetic	86.33%	100.00%
Overlapping synthetic	-0.03%	-6.48%
Wine Quality	0.78%	2.78%

Table 3. Change from random initialization to k-means++.

The separated dataset gives the clearest result. Random initialization sometimes finds a bad local minimum, while k-means++ always finds the good solution in these trials.

The overlapping dataset is a negative result. K-means++ is slightly worse here. This shows that careful initialization helps, but it does not solve every clustering problem.

The Wine Quality result is positive but small. K-means++ has lower mean inertia and slightly lower variability. This is realistic because real data are noisy and not perfectly separated.

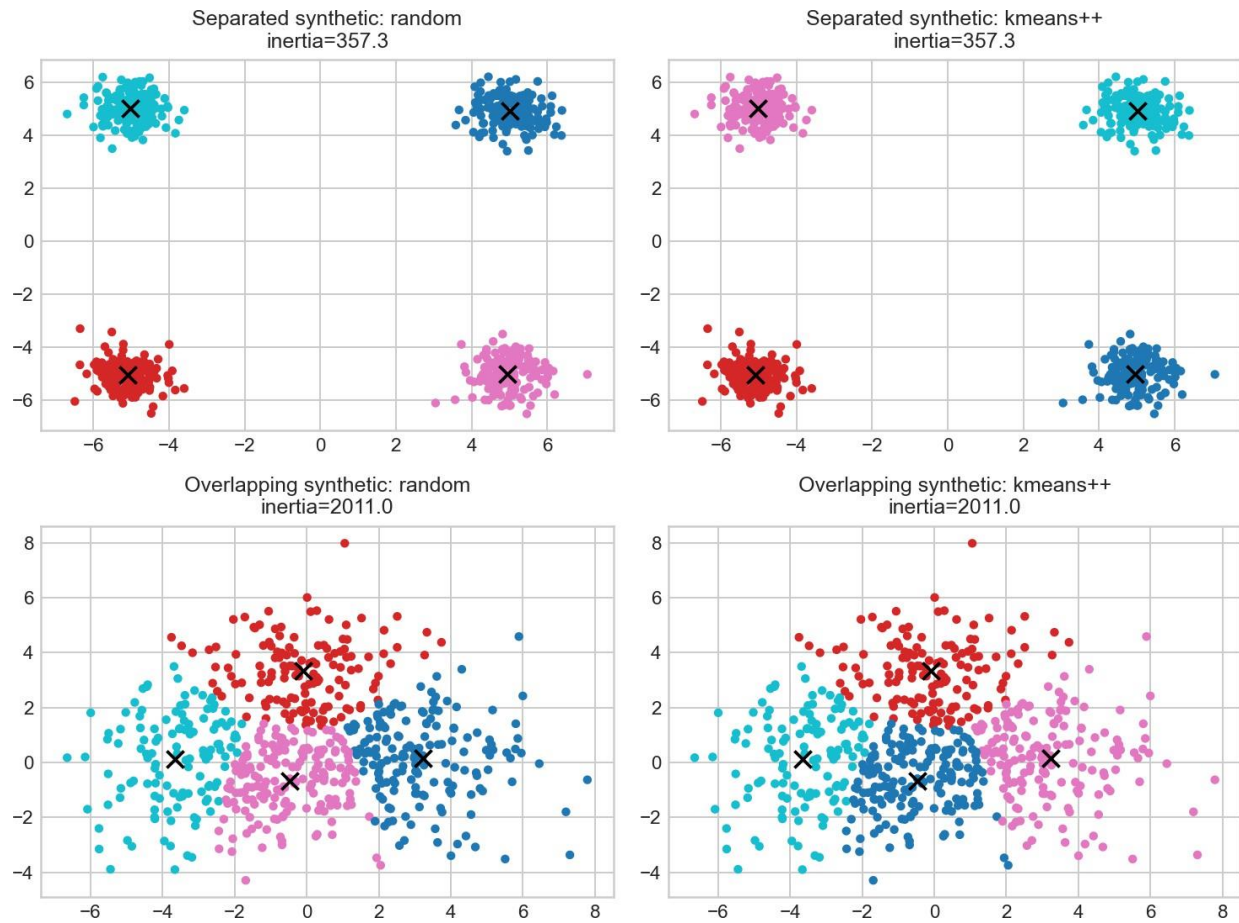


Figure 2. Best synthetic clustering results.

The plots show why the separated data are easy and the overlapping data are difficult. In the overlapping case, several clusterings can look reasonable.

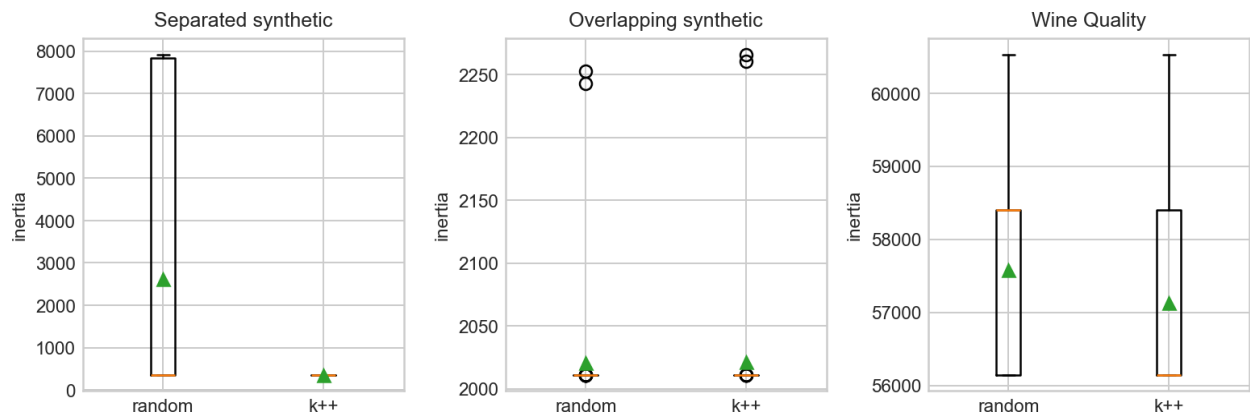


Figure 3. Boxplots of inertia.

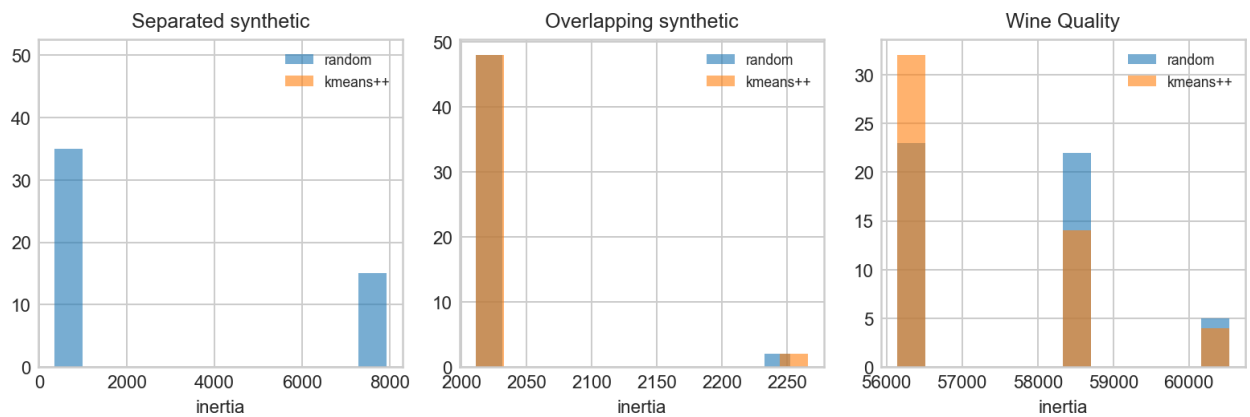


Figure 4. Histograms of inertia.

The distribution plots show stability. K-means++ is much more stable on the separated dataset. For Wine Quality, the distributions are close, but k-means++ is slightly better.

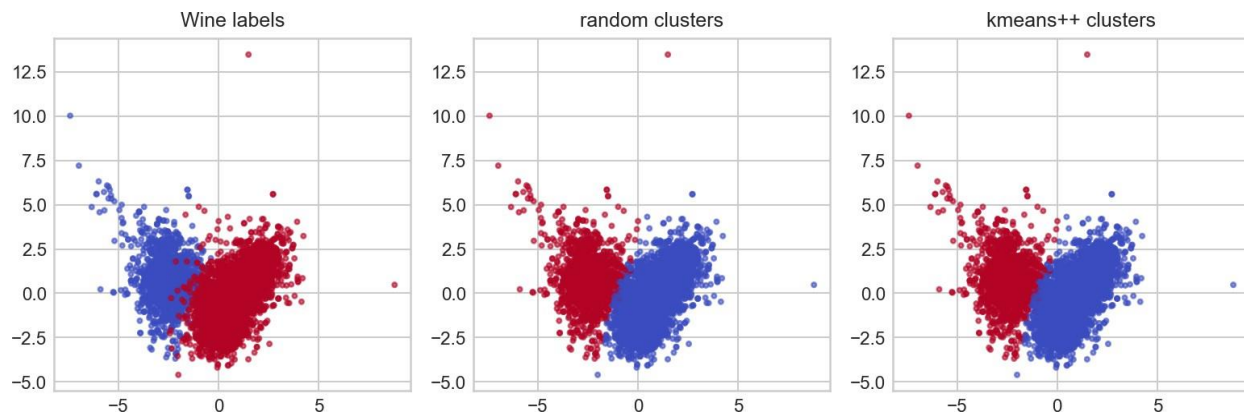


Figure 5. Wine PCA visualization. PCA is only for plotting, not for clustering.

5. Discussion

The positive side is that k-means++ can reduce bad starts. This is very clear on separated clusters and still visible on Wine Quality. The method is simple and only changes the initialization step.

The negative side is that k-means++ is still random and still uses Lloyd's local search after initialization. It does not guarantee the global optimum. The overlapping dataset is the best example in this project: the clusters are mixed, so initialization alone cannot make the problem easy.

High-dimensional real data are also harder than synthetic data. Wine has 11 features, different feature scales, and no perfect spherical clusters. Standardization helps, but Euclidean distance is still only a simple approximation of similarity.

6. Conclusion

This project shows that initialization matters in k-means. K-means++ improves reliability on the separated synthetic dataset and gives a small improvement on Wine Quality. It does not improve the overlapping dataset, which is an important limitation. Overall, the experiments support the idea that careful initialization helps k-means, but it is not a perfect solution.

References

- Arthur, D., and Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding.
- Cortez, P., Cerdeira, A., Almeida, F., Matos, T., and Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties.
- Course lecture materials on clustering and empirical risk minimization.